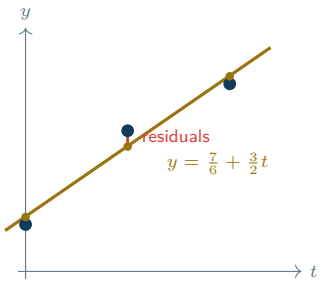
LINEAR ALGEBRA LADDER 9 OF 9

Least squares and orthogonal projection

When $Ax = b$ has no solution, the best answer projects b onto the column space and leaves a perpendicular error.

1 / THE RULE AND THE PICTURE

The best-fit line makes the squared vertical errors as small as possible



Three points $(0,1)$, $(1,3)$, $(2,4)$ do not lie on a line, so $\begin{pmatrix}1&0\\1&1\\1&2\end{pmatrix}\begin{pmatrix}c\\m\end{pmatrix} = \begin{pmatrix}1\\3\\4\end{pmatrix}$ has no solution.

The least squares answer $\hat{x}$ minimises $|b - Ax|^2$ and satisfies the normal equations

$$\boxed{A^T A\,\hat{x} = A^T b}$$

which give the line $y = \frac{7}{6} + \frac{3}{2}t$.

2 / SOLVING IT

The two-by-two normal equations have a unique answer

Here $A^T A = \begin{pmatrix}3&3\\3&5\end{pmatrix}$ and $A^T b = \begin{pmatrix}8\\11\end{pmatrix}$, with $\det A^T A = 15 - 9 = 6$, so

$$\hat{x} = \frac{1}{6}\begin{pmatrix}5&-3\\-3&3\end{pmatrix}\begin{pmatrix}8\\11\end{pmatrix} = \frac{1}{6}\begin{pmatrix}7\\9\end{pmatrix} = \begin{pmatrix}7/6\\3/2\end{pmatrix}.$$

The fitted values are $A\hat{x} = (\frac{7}{6},\ \frac{8}{3},\ \frac{25}{6})$ and the errors $e = b - A\hat{x} = (-\frac{1}{6},\ \frac{1}{3},\ -\frac{1}{6})$, with squared length $\frac{1}{36} + \frac{1}{9} + \frac{1}{36} = \frac{1}{6}$.

3 / THE GEOMETRY / PROJECTION ONTO THE COLUMN SPACE

The error is perpendicular to every column, which is exactly what the normal equations say

Writing $A^T(b - A\hat{x}) = 0$ says the error e is perpendicular to each column of A: dotting with the ones column gives $-\frac{1}{6} + \frac{1}{3} - \frac{1}{6} = 0$ and with $(0,1,2)$ gives $\frac{1}{3} - \frac{1}{3} = 0$.

The fitted vector $A\hat{x}$ is therefore the projection of b onto the column space.

$$P = A(A^T A)^{-1}A^T = \frac{1}{6}\begin{pmatrix}5&2&-1\\2&2&2\\-1&2&5\end{pmatrix}$$

has $P^2 = P$ and $P^T = P$, and $Pb = A\hat{x}$.

4 / WHY THE RESIDUALS SUM TO ZERO

A column of ones among the columns forces it

Because the first column of A is all ones, the equation $A^T e = 0$ contains $\sum e_i = 0$, so the best-fit line passes through the average point: the means are $\bar{t} = 1$ and $\bar{y} = \frac{8}{3}$, and indeed $\frac{7}{6} + \frac{3}{2}\cdot 1 = \frac{8}{3}$.

The matrix $A^T A$ is symmetric with eigenvalues $4 \pm \sqrt{10}$, both positive, which is why it is invertible whenever the columns of A are independent.

ASIDE / ONE IDEA THROUGHOUT

Projection onto a line from the dot product sheet is this problem with one column.

LINKS TO OTHER SHEETS

The projection formula is in *Dot products, length and projection*; the symmetric eigenvalues are in *Eigenvalues and eigenvectors*; the previous sheet is *Diagonalisation and powers*.

Convention. $\hat{x}$ is the least squares solution, $e = b - A\hat{x}$ the error, P the projection onto the column space of A.